Circumgalactic dust halos update #2

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1 Executive Summary

1. Literature review – very recent developments in modeling
2. Accompanying modeling updates (and code improvements)
3. Updated results based on new models – including fitting power laws! The TL;DR is that while the *slope* for all redMaGiC-based XXX samples XXX is a very close match to the Ménard result, the *amplitude* is about twice as high. This is something we can and should look into in future work.
4. For the first time, I have also expand our search for extended dust halos to the Northern Sky! I tried a few different SDSS background samples as well as two different foreground samples: WISExSCOS and GWSLC (itself divided by specific star formation rate). foreground samples. While the WISExSCOS x SDSS results tended to be consistent with the redMaGiC samples at *low* impact parameters, at *high* impact parameters ($>$ XX arcminutes), the extinction flattens out and doesn't decrease. Meanwhile, the GWSLC x SDSS extinction has an interesting dip at intermediate impact parameters but agrees with the redMaGiC curves at high and low impact parameters.
5. Also, my talk at the Aspen Center for Physics dust workshop went really well! There were good questions and a lot of interest afterwards.

2 Literature Review

Since I prepared my last update, several exciting and highly germane papers have been published. Results include an interesting hydrosim study of dust survival in the IGM, a new model for dust extinction, and an update to the canonical SFD map.

2.1 Gordon et al. 2023

described in [2]. Implemented with the `dust_extinction` software package [1].

2.2 Helena outflow paper

2.3 CSFD

2.4 Dust as a supernova systematic

3 Code and Modeling Updates

3.1 Code updates

3.1.1 Color-matched randoms

- describe algorithm, show nice figure
- This didn't significantly affect the result, however.

3.1.2 Random catalog generation overhaul

- Mention bug of picking healpixels; could even show the bad ra/good ra figures
- Also implemented ability to generate catalogs in galactic coordinates, which made sense for the WISExSCOS catalog
- Based on the redMaGiC ratio of random galaxies to real galaxies, the random catalogs I create are $\sim 20\times$ the length of their respective data catalogs. For very large galaxy catalogs like SDSS, generating an equivalent random catalog with almost a billion objects is too much for my HPC to handle. (Or, perhaps my code is inefficient.) So, I added a little option to create multiple smaller catalogs and concatenated them.

3.1.3 Treecorr variance and misc. utilities

- Fixed variance! Used patches and compensated estimates
- `RADec_to_healpixel`
- deredden magnitudes, the script used to incorporate CSFD extinction values into magnitudes.
- fix magnitude (idnr what that is)

3.2 Model improvements

3.2.1 Gordon 2023 extinction

- At the workshop, Daniella Calzetti told me that she was flattered but that I needed to update the extinction law that I was using. She pointed me to the Gordon 2023 paper referenced above.
- I found the `dust_extinction` repository[1] and used that.
- It did shift our values, but the changes are irrelevant since the previous values were so out of date!

[1] https://github.com/karllark/dust_extinction

3.2.2 CSFD

- As metioned earlier, Nao pointed me to this CSFD paper, where the amplitude of the residual seemed non-negligible (10^4) and so worth implementing.
- I located a nice software package called `dustmapper`[2] to query the CSFD dust map and obtain extinction coefficients for a given set of coordinates. Specifically, there is a handy `dustmaps.csfd.CSFDQuery` method that takes as input an RA, Dec list in SkyCoords format and returns an array of E(B-V) values. An R_V value is used to convert those into extinction coefficients and then used to de-redden SDSS and DES magnitudes. Using SFD maps to obtain E(B-V) Applying band-specific extinction coefficients to get extinction values.
- Subtracting these extinction values $A_{\mathrm{X,CSFD}}$ from the observed magnitudes `MAG_X` to obtain the dereddened magnitudes:

  `MAG_X_CORRECTED = MAG_X - A_X_CSFD`

  For the DES magnitudes, I used the corrected magnitude prescription of Pandey et al. 2021:

  `MAG_CORRECTED = MAG + DELTA_MAG_Y4 + DELTA_MAG_CHROM - A_X_CSFD`
- As one might expect, this had a notable effect change

4 Catalogs: Round 2

5 Results

6 Discussion and Next Steps

References

[1] Karl D. Gordon. "dust_extinction: Interstellar Dust Extinction Models". In: *Journal of Open Source Software* 9.100 (2024), p. 7023. DOI: `10.21105/joss.07023`. URL: `https://doi.org/10.21105/joss.07023`.

[2] Karl D. Gordon et al. "One Relation for All Wavelengths: The Far-ultraviolet to Mid-infrared Milky Way Spectroscopic R(V)-dependent Dust Extinction Relationship". In: *Astrophysical Journal* 950.2, 86 (June 2023), p. 86. DOI: `10.3847/1538-4357/accb59`. arXiv: `2304.01991 [astro-ph.GA]`.

[2] `https://github.com/gregreen/dustmaps`